RESEARCH PROPOSAL · LANGUAGE FUTURES PROGRAMME 2027

Qillqa

Community-Grounded Speech and Text Technology for Southern Quechua and Aymara

PRINCIPAL INVESTIGATOR

Dr Inés Condori Huamán

DURATION

36 months, from 1 April 2027

AMOUNT REQUESTED

EUR 1,060,000

HOST INSTITUTION

Andes Institute for Language and Computation, Cusco

SUMMARY

Southern Quechua and Aymara are spoken by more than nine million people, yet neither has a speech recogniser, spell checker or machine translation system that its speakers would choose to use. Qillqa will build these tools together with three speaker communities in Cusco, Puno and La Paz. Over three years we will record and transcribe 600 hours of consented speech, release an open text corpus under community data governance, adapt multilingual models to both languages with less than a tenth of the data such models usually require, and evaluate every system with speakers rather than with automatic scores alone. The project ends with tools that run offline on an ordinary phone, and with a governance model that other indigenous language communities can adopt.

1 Background

Language technology now reaches most of the world's speakers, but not most of the world's languages. Of the roughly 7,000 languages in use, fewer than a hundred have the text and speech resources on which current systems are trained (Ortega et al. 2024). Quechua and Aymara are large by any demographic measure (Adelaar and Muysken 2004), yet they sit firmly on the wrong side of this divide: the largest public Quechua text collection holds under four million words, much of it translated religious material (Huamán et al. 2023).

Two recent developments make progress possible. First, massively multilingual models transfer surprisingly well to languages they have barely seen, provided the target language shares script and some vocabulary with the training mixture (Lindqvist et al. 2025). Second, community-led data initiatives have shown that speakers will

contribute recordings in large numbers when they keep control over how the data is used (Carroll et al. 2020; Tupa Mamani et al. 2022). What is missing is a project that joins the two: one that treats speakers as partners in design and evaluation, not only as a source of data.

Evaluation is the weakest link. Automatic metrics such as word error rate say little about whether a transcription is usable, and they are unreliable for agglutinative languages, where a single morpheme error changes a whole word (Mendoza and Fischer 2024). Qillqa therefore makes human evaluation by speakers a first-class part of the work plan (Section 4).

2 Aims

The project pursues three aims. Each is tied to one or more work packages in Figure 1, and each has objectives that a reviewer can check at the end of the grant.

> AIM 1
>
> Build open, consented resources
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> Create speech and text collections for both languages under a data governance framework that the partner communities write and enforce.
>
> - O1.1 Sign data agreements with three community organisations before any recording starts.
> - O1.2 Record and transcribe 600 hours of speech across six dialect areas.
> - O1.3 Release a 25-million-word text corpus with per-document provenance.

> AIM 2
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> Adapt models with little data
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> Show that multilingual models can reach usable quality in Quechua and Aymara with a fraction of the data they normally need.
>
> - O2.1 Reach a word error rate below 18 % on conversational speech.
> - O2.2 Match a strong supervised baseline in translation to and from Spanish using under 200,000 sentence pairs.
> - O2.3 Distil every model to run offline on a phone with 4 GB of memory.

> AIM 3
>
> Evaluate with speakers
>
> Replace automatic scores as the final judge with structured evaluation by speakers of each dialect, and publish the protocol.
>
> - O3.1 Run four rounds of speaker evaluation, each with at least 60 participants.
> - O3.2 Publish an evaluation protocol that other projects can reuse.

3 Methods

3.1 Data collection and governance

Recording follows the protocol of Tupa Mamani et al. (2022): speakers choose the topics, keep the right to withdraw their recordings, and decide together which uses are permitted. Transcription is done by trained community research assistants, paid at the regional rate for skilled work, with a second assistant checking a random 10 % sample. No recording is collected before Objective O1.1 is met.

3.2 Model adaptation

We start from an open multilingual speech encoder and an open translation model, and adapt both in two stages. The first stage continues pre-training on untranscribed audio and monolingual text. The second fine-tunes on labelled data with a loss that weights each dialect d by the inverse of its share p_d of the training set, so that minority dialects are not drowned out:

$$\mathcal{L} = \sum_{d \in D} \frac{p_d^{-\alpha}}{\sum_{d'} p_{d'}^{-\alpha}} \, \mathbb{E}_{(x,y)\sim \mathcal{D}_d}\left[-\log p_\theta(y \mid x)\right], \tag{1}$$

where $\alpha \in [0, 1]$ controls how strongly rare dialects are favoured (Arivazhagan et al. 2019). We tune α on held-out speakers, never on held-out sentences from speakers seen in training.

3.3 Evaluation

Each evaluation round pairs automatic metrics with a speaker study. Speakers rate transcriptions and translations for adequacy and naturalness on a five-point scale, and mark errors that would change the meaning. We report agreement between raters and between ratings and Equation (1)-trained model scores, following Mendoza and Fischer (2024).

4 Work plan

The work is organised in five work packages over twelve quarters (Figure 1). Governance runs through the whole project, because data agreements are revisited at every release.

Table 1 Milestones and the quarter in which each is due.

MILESTONE	DELIVERABLE	DUE
M1	Data agreements signed with all three partner communities	Q2
M2	First public release of the speech and text corpus	Q6
M3	Adapted speech and translation models reach the targets of Aim 2	Q8
M4	Fourth speaker evaluation round completed and analysed	Q11
M5	Offline tools, corpus and evaluation protocol released	Q12

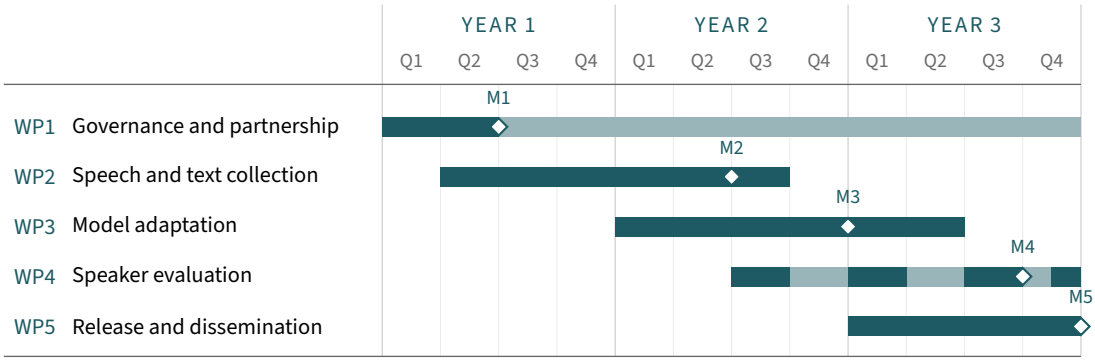


Figure 1 Work plan by quarter. Dark bars mark intensive work and pale bars continuing work; in WP4 the dark blocks are the four speaker evaluation rounds. Diamonds mark the milestones listed in Table 1.

5 Risks and mitigation

Table 2 lists the risks we consider most likely to delay the project, with the step we take before each one occurs and the fallback if it does.

Table 2 Main risks, their assessed likelihood and impact, and how each is mitigated.

RISK	LIKELIHOOD	IMPACT	MITIGATION
Data agreements take longer than planned	Medium	High	Negotiation starts six months before the grant; WP3 can begin on existing public text.
Too few speakers in one dialect area	Medium	Medium	Recruit through radio stations and schools; merge with a neighbouring area after Q4 if needed.
Models miss the error-rate target	Low	High	Report the gap openly; shift effort from translation to speech, where gains are largest.
Loss of a community research assistant	Medium	Low	Train two assistants per site, so that each site keeps one trained member at all times.
Compute costs rise during the project	Low	Medium	Budget includes a 15 % contingency on compute; distilled models reduce later training cost.

6 Budget

The amount requested is EUR 1,060,000 over three years (Table 3). Personnel makes up 74 % of direct costs; most of the rest pays speakers and community workshops, which we treat as part of the research, not as overhead.

Table 3 Budget by year, in euros. Indirect costs are the programme's flat rate of 25 % of direct costs.

COST CATEGORY	YEAR 1	YEAR 2	YEAR 3	TOTAL
Postdoctoral researchers (2.0 FTE)	118,400	121,900	125,600	365,900
Doctoral researcher (1.0 FTE)	38,200	39,300	40,500	118,000
Community research assistants	46,800	48,200	49,600	144,600
Speaker compensation and workshops	22,500	28,000	18,500	69,000
Recording equipment	14,800	2,000	0	16,800
Travel and fieldwork	18,600	21,400	16,200	56,200
Compute and storage	9,500	24,000	21,500	55,000
Dissemination and open access	2,000	6,500	14,000	22,500
Direct costs	270,800	291,300	285,900	848,000
Indirect costs (25 %)	67,700	72,825	71,475	212,000
Total requested	338,500	364,125	357,375	1,060,000

7 Expected impact

For speakers, Qillqa delivers a speech recogniser, a spell checker and a translation aid that work without a network connection, in the dialects people actually speak. For the field, it provides the first evaluation protocol for these languages that is grounded in speaker judgement, and evidence on how far multilingual transfer can be pushed with community-scale data (Nakamura et al. 2026). For other indigenous language communities, the governance agreements and their record of use over three years offer a tested model, in the spirit of the CARE principles (Carroll et al. 2020), that can be adopted without starting from nothing. All software is released under an open licence; the data is released under the terms each community sets in Aim 1.

8 References

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